

Case management for high-intensity service users: towards a relational approach to care co-ordination

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What is known about this topic

- Case managers are at the forefront of attempts to develop services for high intensity service users.
- They can act as effective care facilitators and resource negotiators.
- Their ability to work effectively is influenced by resource constraints, service ideology and the quality of organizational infrastructure.

What this paper adds

- The pattern of interaction between case managers and their co-workers influences the scope and quality of care co-ordination.
- Clearly defined, relational workspaces help case managers to engage with their co-workers and support the management of inter-dependent care activities.
- Flexible and efficient, task focussed workspaces are needed to improve the quality of care co-ordination in dispersed networks of care.

Abstract

This study is based on a formative evaluation of a case management service for high-intensity service users in Northern England. The evaluation had three main purposes: (i) to assess the quality of the organisational infrastructure; (ii) to obtain a better understanding of the key influences that played a role in shaping the development of the service; and (iii) to identify potential changes in practice that may help to improve the quality of service provision. The evaluation was informed by Gittel's relational co-ordination theory, which focuses upon cross-boundary working practices that facilitate task integration. The Assessment of Chronic Illness Care Survey was used to assess the organisational infrastructure and qualitative interviews with front line staff were conducted to explore the key influences that shaped the development of the service. A high level of strategic commitment and political support for integrated working was identified. However, the quality of care co-ordination was variable. The most prominent operational factor that appeared to influence the scope and quality of care co-ordination was the pattern of interaction between the case managers and their co-workers. The co-ordination of patient care was much more effective in integrated co-ordination networks. Key features included clearly defined, task focussed, relational workspaces with interactive forums where case managers could engage with co-workers in discussions about the management of interdependent care activities. In dispersed co-ordination networks with fewer relational workspaces, the case managers struggled to work as effectively. The evaluation concluded that the creation of flexible and efficient task focused relational workspaces that are systemically managed and adequately resourced could help to improve the quality of care co-ordination, particularly in dispersed networks.

Keywords: care co-ordination, case management, dispersed networks, high-intensity service users, integrated networks, relational approach

Introduction

The growing number of frail older people with multiple, long-term conditions such as diabetes, heart failure, chronic respiratory problems and depression presents a major challenge to public health and social services

across the world (Strong *et al.* 2005, Mathers & Loncar 2006). Evidence suggests that the systemic implementation of integrated, patient centred, case management approaches such as the Chronic Care Model (Wagner *et al.* 1996, 2001, Bodenheimer *et al.* 2002), can help to maximise the health status, level of functioning and qual-

ity of life of this vulnerable population (Ouwens *et al.* 2005, Minkman *et al.* 2007, Parchman *et al.* 2007). Case management generally involves a combination of core elements including: assessment, planning, linkage, monitoring, advocacy and outreach (Huber 2002). However, there is a considerable variation in case management models that are adopted, particularly in local contexts and the extent to which case managers provide direct clinical care, psychosocial support and responses to crisis situations (Sargent *et al.* 2007).

Under optimal conditions, case managers for high-intensity service users should act as effective care facilitators and 'resource negotiators' (Zander 1995), assisting service users to manage their conditions, deal with exacerbations, cope with transfers of care between different care providers and deal with the everyday stresses of living with chronic illness (Lynn 2000, Kodner & Spreeuwenberg 2002, Chapman *et al.* 2009). Yet, their ability to carry out their role effectively may vary, due to differences in the resources available, service ideology and the quality of the local organisational infrastructure (Hroschikoski *et al.* 2006, Iliffe 2006, Sargent *et al.* 2008). The national evaluation of the Evercare approach towards case management in England identified that the intervention had no significant effect on the level of emergency admissions and concluded that more radical service redesign was needed to improve outcomes (Gravelle *et al.* 2007). Given that case managers are currently at the forefront of attempts to develop integrated services for high-intensity service users within the English National Health Service (Department of Health 2004, 2005a, 2006), there is an urgent need to develop a better understanding of influences that may enable or constrain the development of case management services (Dixon 2007).

This study outlines the findings from a formative evaluation of a case management service for high-intensity service users run by the provider arm of a Primary Care Trust in Northern England. The evaluation had three main purposes: first, to assess the quality of the organisational infrastructure supporting the case management service; second, to try to obtain an understanding of the key influences that played a role in shaping the development of the service; and third, to identify potential changes in practice that may help to improve the quality of service provision.

Theoretical framework

The evaluation was informed by Gittell's relational co-ordination theory, which focuses upon cross-boundary working and the social relationships that facilitate task integration (Gittell 2002a,b, Gittell & Weiss 2004). The origins of Gittell's approach are rooted within organisa-

tional sociology and the study of high-performance work systems (Hackman 1987, Pfeffer 1998). Gittell examines the dynamic, interactive processes that shape collective working practices and explores how human resources can be used in productive and efficient ways. In doing so, she has developed an alternative to bureaucratic based theories, which frame co-ordination as a function of impersonal, instrumental control (Pfeffer & Salancik 1978) and network theories which map the influence of social ties and attitudes upon general behaviour (Kilduff & Brass 2010). Gittell conceptualises co-ordination as the management of task interdependencies via relational networks. High quality co-ordination is facilitated by social bonds and shared identities, which are underpinned by three relational mechanisms: shared goals, shared knowledge and mutual respect. Relational boundary spanners such as case managers can build bridges via these mediating mechanisms and improve task integration (Gittell 2002a, 2003, 2009).

Support for Gittell's theory has come from empirical studies that have identified the impact of different communication patterns and relational networks between key service providers. A study of surgical units in nine hospitals, for example, identified a clear correlation between the quality of relational co-ordination and overall quality of care provision, as measured by length of stay, levels of postoperative pain and patient satisfaction with the quality of care (Gittell *et al.* 2000). The study found that relational co-ordination was improved by the development of a shared ethos, as well as formal working practices and functional routines that facilitated frequent, timely communication, joint problem-solving and mutual respect. This type of relational co-ordination is necessary to facilitate efficient and effective task integration in conditions characterised by complex task interdependencies, clinical uncertainty and role ambiguity (Gittell *et al.* 2008). Gittell's theory may therefore be especially relevant to the co-ordination of care for high-intensity service users, in which a combination of service providers is typically required to act in a timely, flexible and integrated way to deal with unpredictable contingencies.

Methods

The evaluation was conducted between February 2007 and August 2008. There were a number of staffing changes during this time period but for the majority of the time that the evaluation was being conducted the service comprised the following staff members; a service manager, three community matrons who acted as both advanced clinical practitioners and case managers, six case managers and an administrator. The service users that the case managers supported had complex needs

arising from conditions that severely compromised their health and ability to function. All had a Patients at Risk of Rehospitalisation (PARR) score of over forty, indicating that there was a high risk they would require emergency readmission to hospital (Billings *et al.* 2006).

The quality of the organisational infrastructure supporting the case management service was measured using the Assessment of Chronic Illness Care (ACIC) Survey (Bonomi *et al.* 2002), a self assessment questionnaire and quality improvement tool which is designed to measure fidelity to the principles of the Chronic Care Model. The ACIC has six subscales (quality of delivery system design, patient-provider relationships, decision support, clinical information systems, community resources and healthcare organisation) as well as an extra subscale that measures the overall integration of the Chronic Care Model. The ACIC questionnaires were posted to 23 of the team members and managerial staff who had been employed or involved in the development of the case management service. Ten questionnaires were completed and returned (response rate 44%). Six were from team members within the case management service and four from managers within the Primary Care Trust.

Semi-structured qualitative interviews were conducted to explore how case managers, community matrons and the managers who were responsible for the service conceptualised their role, their position within networks of care and their responses to the challenges they faced. The interviews took the form of a guided conversation (Rubin & Rubin 1995) based on a topic guide which was developed after three pilot interviews (see Figure 1). Potential interviewees were approached using formal channels such as staff meetings and informal contacts that were developed with the practitioners who worked for the service. Everyone who was approached agreed to be interviewed. The interviewees were given

an assurance of anonymity and all of the interviewees agreed to have their interviews digitally recorded. A total of 19 individual and four group interviews were conducted. The first nine individual and two group interviews were conducted with a convenience sample. The remainder of the interviews were conducted with a purposively selected sample of interviewees who were identified as key informants who were able to expand upon the key themes that were emerging from the analysis. The average duration of the interviews was approximately 45 minutes. The interviews were conducted in the interviewees' workplace over the course of a 12-month period and the topic guide was adapted to incorporate emerging themes as the interviews progressed. Five of the interviewees were interviewed on more than one occasion. The interview data were analysed using a combination of elements from framework analysis (Ritchie & Spencer 1994) and the realistic evaluation approach developed by Pawson & Tilley (1997). The interviews were independently analysed by two of the authors (PM and DE) who then worked together to code the transcripts and construct a thematic matrix. Five key mechanisms that were central to the case manager's role were abstracted, plus one key contextual factor: the patterns of interaction between the case managers and their co-workers (see Figure 2). This matrix was then used to explore the links between the themes and facilitate the interpretation of the dataset as a whole.

Findings

Organisational infrastructure

The scores on the ACIC subscales are presented in Figure 3. The results indicate that the area perceived as

Role of case manager ▪ How would you describe your role as a case manager? ▪ What type of work does this involve? Factors influencing work ▪ What sorts of things help you to do your work well? ▪ What sorts of things get in the way? Working relationships ▪ How would you describe, both the nature and the quality of, your relationship with the other service providers you work with? Internal organisation/Team working ▪ Can you tell me about the internal organization of the service and how you work as a team? Other issues ▪ Are there any other issues that you would like to comment on?
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Figure 1 Interview guide.

- Boundary spanning – forging links and connections across service boundaries
- Resource negotiation – discussions with the institutional gatekeepers who exercise control over resources via the interpretation of eligibility criteria and negotiations about the mechanics of care ‘hand overs’
- Heedful interacting – responding in a prompt and flexible way to deal appropriately with changes in circumstances
- Functional routines – norms and procedures designed to facilitate co-working
- Performance measurement – the process of quantifying efficiency and effectiveness
- The patterns of interaction between the case managers and their co-workers – the pattern of communication and exchange relationships with members of different disciplines, organizations and occupational communities who needed to work together in order to facilitate task integration

Figure 2 Items in thematic matrix.

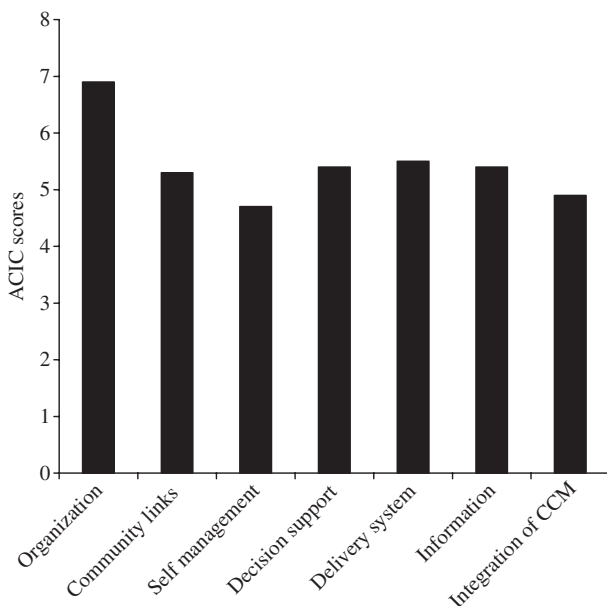


Figure 3 Results from Assessment of Chronic Illness Care (ACIC) Survey.

most developed was the organisation of the healthcare system. This subscale represents the perceived level of strategic commitment and political support from key organisational leaders. The scores on the ACIC subscales that measured the quality of care processes such as self management support, community linkages and overall integration of the Chronic Care Model were substantially lower.

The interview data confirmed that the development of case management and integrated working was a strategic priority for the Primary Care Trust and Local Authority. This was reflected in comments regarding

the substantial investment that had been made in developing integrated care systems. These systems had joined up organisational structures, standardised assessment procedures, single points of access and co-located multi-disciplinary teams with integrated information systems.

Influences on service development

It was clear that case managers faced considerable challenges, as they endeavoured to establish and maintain an effective service that improved the quality of care co-ordination and collaborative working. Many service users lived in exacting social circumstances, alone or with carers who were struggling to cope and their difficulties were regularly compounded by alcohol abuse, mental health problems and cognitive decline. Furthermore, their relationships with service providers were sometimes ambivalent, as they struggled to retain a sense of stability and control over the course of their lives.

Newly appointed case managers, in particular, highlighted difficulties that they encountered in adjusting to the demands of their role. For example, a case manager who had previously worked for many years as a district nurse, described how she had been accustomed to delivering discrete, time limited interventions for service users with conditions such as chronic obstructive pulmonary disease:

As a district nurse, I did ask you know, were they okay with their inhalers, could they show me their technique. But that would be my only involvement. Whereas now when I'm at some of these patients' houses and they've got four inhalers and I'm having to sort of do an idiots' guide. What I'm sadly lacking in is confidence. That might be an age thing. I seem to be trying to pick things up as I go along.

An equal, if not much greater challenge than that of developing the requisite technical knowledge and clinical skills, was the task of co-ordinating patient care. Three relational mechanisms were identified that were central to the co-ordination of care by the case managers; boundary spanning, resource negotiation and heedful interaction. Boundary spanning entailed forging links and connections with co-workers and other agencies that transcended the boundaries of the service. Resource negotiation involved discussions with gatekeepers who exercised control over access to information, expertise and material resources, concerning issues such as the application of eligibility criteria and the mechanics of transfers of care. Heedful interactions involve responses that are prompt, flexible and cohesive and that deal appropriately with changes in circumstances and unexpected contingencies (Weick & Roberts 1994). The way in which the case managers sought to co-ordinate patient care via these mechanisms was influenced by the patterns of communication and interaction with their co-workers, together with the characteristics of the networks they accessed. There was a continuum of network structures ranging from formally organised, integrated co-ordination networks to dispersed co-ordination networks. The defining difference between the network types was the pattern of interaction and the quality of the case managers' interpersonal relationships with their co-workers. Integrated networks were characterised by a sense of common purpose. They were supported by regular, informal exchanges and the presence of relational workspaces, interactive forums that facilitated task focused discussions about interdependent care activities. Dispersed networks were characterised by more temporary links and patterns of interaction. Joint working occurred in dispersed networks, however, connections between co-workers were not grounded by the same sense of collective consciousness. Both network types were located within and across formal organisational boundaries.

Integrated networks

Integrated networks were built upon relatively strong relational ties and structured organisational systems that enabled the transfer of information and the leveraging of resources. The institutionalised norms and procedures in these networks facilitated collaborative, interdisciplinary working via integrated care pathways and programmes for the delivery of population based chronic illness care. A structured, collegial atmosphere was sustained by regular, ad hoc exchanges between team members and their joint participation in organisational team meetings. This helped to foster a sense of collective responsibility. Team members engaged in co-operative problem-solving and worked towards common goals. Working in co-located

integrated nursing and social care teams, with one point of entry, shared assessment and information systems and a single team manager, made it far easier for case managers to harness the collective effort of team members and access discrete resources. A case manager in an integrated health and social care team described how integrated working had helped her to secure access to sheltered accommodation for a service user.

On the board you can see at a glance who is involved with your patients and if you're sat next to them (social worker in shared office) they don't pass it back. They know what they're doing there's no wait. They know who to ring, how the system works.

Dispersed networks

There was a diverse array of statutory and non-statutory care providers who were connected to the case managers through weak ties (Granovetter 1973) and loose institutional arrangements. These care providers included specialist hospital-based services, nursing homes, intermediate care facilities, community nursing, mental health and social services. The transfer of information, the leveraging of resources and the rationale for joint working in the dispersed networks tended to be more ad hoc and were often negotiated between co-workers on a case-by-case basis. There were relatively few standardised routines to facilitate collaborative working. Care procedures were often difficult to enact due to an absence of relational workspaces and the difficulty of combining the work of services whose operations were shaped by divergent norms, expectations, organisational processes and temporal rhythms. Boundary spanning, resource negotiation and heedful responses were more time consuming within the dispersed networks and connections with co-workers were more difficult to maintain.

It was usually possible for the case managers to negotiate access to resources to deal with cases of exceptional need. However, when dealing with non-urgent matters, case managers had to be much more resourceful and creative in using their local knowledge, contacts and negotiation skills to secure the co-operation of co-collaborators and organisational gatekeepers. Care providers often exercised their power of veto and withheld their co-operation. For example, a case manager described the 'brick walls' she had encountered when she had tried to act as an advocate for a service user who risked losing her home.

I've rung the social worker, housing, welfare rights they're not interested. They all said it's nothing to do with us. Duty (social services) told me that they can't take it on.

Some of the case managers were especially skilled at breaking down boundaries and cultivating relationships

across dispersed networks. An illustrative example was provided by a case manager who described how she had secured the co-operation of a General Practitioner who had initially opposed the introduction of the case management service.

There was one ... we didn't hit it off at all but we've made massive inroads and now it's 'Come and sit down' and you know she gets her pictures of her baby out. We've broken the ice and she will come to me and say 'Will you come in and discuss these patients?'

However, faced with practical difficulties of trying to co-ordinate activities across networks that were characterised by ambiguous and overlapping spheres of responsibility, poor communication and contested role boundaries, the principal response of the case managers was often to try to muddle through and steer clear of conflict. This approach was typified by a case manager who explained how she had become cynical about the prospect of ever developing a collaborative relationship with some primary care practices.

I go into Practices on a regular basis, less so in the one where relationships aren't established because I feel that I'm wasting my time.

It was not uncommon for frustrated case managers to deal (more or less exclusively) with the problems that they could address themselves. In order to avoid paralysing debates about whether or not service users met the eligibility criteria for specific health or social services, case managers often sought to work around the system and fill in gaps in service provision themselves. Although this 'hands on' work helped to reduce the duplication of services, the downside of this work was that it was very time consuming and left them with less time to concentrate on the co-ordination of care. Rather than acting as boundary spanners and system integrators, they then delivered interventions within the confines of their own individual roles. The knock on effect of this dynamic was that it impacted upon the ability of the case managers to manage their heavy workloads. Keeping track of and maintaining contact with all of the service users on their caseloads became increasingly difficult. A corollary of this was that it was common for case managers to have a number of residual 'sleepers', who received very little active input. This situation was viewed as highly unsatisfactory. One case manager commented:

You're just another person who's gone in and said 'Oh I could help with that' and done one.

These pressures were exacerbated by the services' performance measurement regime. The process of quantifying the efficiency and effectiveness of the service gave

rise to three related challenges. First, pressure was being placed upon the service managers, by commissioners who were seeking to ensure that the case managers achieved the caseload targets of 50–80 service users specified in guidelines issued by the Department of Health (2005b). Second, a large amount of time was absorbed completing mandatory paperwork. Third, the significant amount of extra time and resources that needed to be invested in establishing relationships and co-ordinating the activities across dispersed co-ordination networks was not formally recognised or acknowledged.

Overall responsibility for caseload management and care co-ordination was left to individual case managers to determine. This was achieved on the basis of their professional judgement only, as they were given no clear guidance or sense of direction. When they were pushed to their limits, the shortcomings of their individualistic approaches were exposed. As one case manager frankly admitted:

We all have our own systems.

What these individual systems and ways of working tended to have in common was that they were based on the pragmatic logic of making do (March & Simon 1958). Operating in this dispersed, fragmented marketplace for scarce resources, it was all too easy for case managers to lose sight of one of their key tasks, which was to co-ordinate the care of the vulnerable, high-intensity service users, and reduce their over reliance on unscheduled and emergency sources of care and support.

Discussion

The ACIC survey results and the data from the interviews with front line staff were consistent with the body of research that has highlighted the benefits of service integration (Johri *et al.* 2003), along with considerable challenges associated with collaborative working and its potential limitations (Glasby & Peck 2005, Glendinning *et al.* 2005, Gravelle *et al.* 2007). A high level of political support for integrated working was identified, together with a considerable amount of investment in interagency working, which was exemplified by the establishment of integrated health and social care teams. There were clear signs that the investment had helped to improve the quality of interdisciplinary working, but despite this investment, there were still significant variations in the quality of care co-ordination.

Factors that may have contributed to the variations in the quality of care co-ordination included the political process of negotiating and defining common interests (Carlile 2004), the local barriers between fragmented health and social care resources and the case managers'

struggle for recognition and status as an emerging occupational group. However, the most prominent operational factor appeared to be the patterns of interaction between the case managers and their co-workers. The co-ordination of patient care appeared to be much more effective in integrated co-ordination networks with clearly defined task focussed relational workspaces and interactive forums where case managers could engage with co-workers in discussions about the management of interdependent care activities. In dispersed co-ordination networks with few established relational workspaces, the case managers struggled to work as effectively. Constrained by the quality of their relationships with other service providers and a relatively impoverished marketplace for resources, they tended to avoid conflict and muddle through. In many cases this involved taking over direct responsibility for care provision themselves. These links are summarised in the context, mechanism, outcome configurations CMO – 1 and CMO – 2 (see Figures 4 and 5).

We may need to re-appraise the role that relational workspaces and team working play in supporting the case management of high-intensity service users, given that high-quality connections between co-workers (Coiera 2000) are required to deal with the shifting and unpredictable contingencies they encounter. The ability to develop and sustain cordial and effective working relationships with their colleagues is among the core competencies that case managers have (Department of Health, 2006). However, they may also need to develop

more specific sets of skills if they are to engage effectively in the relational mechanics of boundary spanning, resource negotiation and the facilitation of heedful responses. For example, advanced entrepreneurial and conflict resolution skills may be needed to disentangle and effectively deal with power differentials and conflicts of interest that can arise when managing complex interdependencies (Gittel 2009). These sorts of skills can be developed through high-quality supervision and training (Carmeli & Gittel 2009). Nevertheless, without a clear mandate it is still unlikely that individual case managers will make a substantial impact in binding fractured organisational relationships (Ring & Van de Ven 1994). Structurally engrained patterns of negative interaction and skewed exchange relationships (Cook 1977) are bound to place constraints on their ability to collaborate with other care providers and it is necessary to develop process level solutions to deal with these challenges (Gittel 2009).

From a relational perspective, structural solutions such as clearly defined care pathways and operational procedures need to be combined with a relational architecture that establishes the integrative conditions that are necessary for effective communication and co-ordination (Okhuysen & Bechky 2009). This relational architecture may need to contain both formal and informal elements, if it is to promote a sense of collective efficacy (Johnson 2003). Informal task focussed discussions can emerge spontaneously when co-workers work in close proximity to each other and the value of these informal interactions

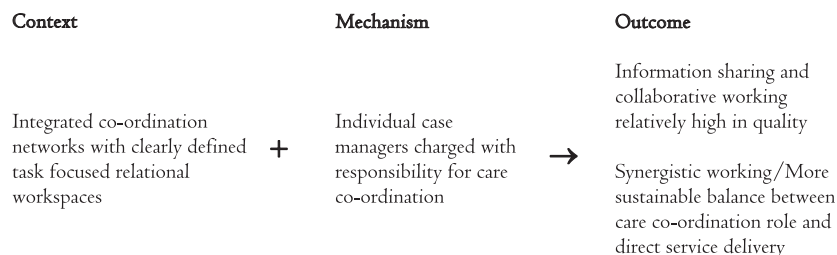


Figure 4 CMO – 1: context, mechanism, outcome configuration for integrated co-ordination networks.

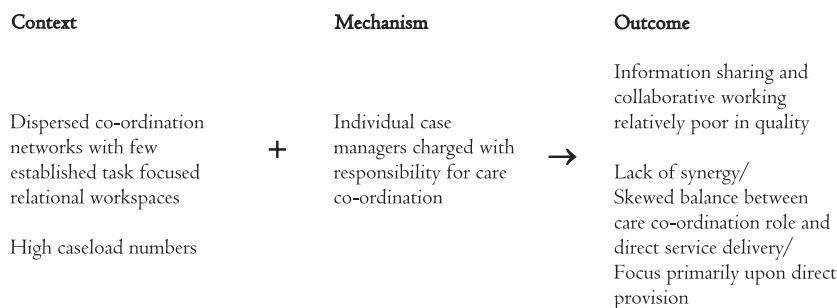


Figure 5 CMO – 2: context, mechanism, outcome configuration for dispersed co-ordination networks.

should not be under-estimated. However, they need to be supported by formal, scheduled meetings and interactive routines, as they are dependent upon personal relationships that are likely to break down due to staff turnover and competing pressures.

In dispersed networks of care, it is not feasible or necessary to engage with all co-workers at the same level of intensity and trade offs have to be made between the numerous demands on case managers' time. Establishing relational priorities, creating functional routines and making judicious use of alternative communication media can enable productive use of time and effort. Thompson's (1967) hierarchical distinction between pooled (separate), sequential and reciprocal interdependence may prove useful, as it could help case managers map out their networks of interdependency. Thompson's tiers of interdependence mirror the different tiers in stepped care approaches to treatment, except that they identify the resources that need to be invested in developing bridging social capital (Adler & Kwon 2002) with co-workers. More sophisticated tools such as process mapping techniques (Savory & Olson 2001) and dependency structure matrices (Browning 2002), can also help to identify critical junctures such as hand overs of care where it is vital to have interactive discussions (Coleman 2003). Functional routines based on modular systems of working (Sanchez & Mahoney 1996) can reduce the need for protracted reflection and provide a stable platform on which to respond to changing circumstances ((Becker 2004, Greenhalgh 2008). Technological approaches such as tele-conferencing can also help case managers to overcome barriers to relational engagement, by establishing the connections and bonds that are envisaged by the advocates of 'virtual teamwork' (Lipnack & Stamps 1997).

Overstretched case managers may resort to the defensive mechanism of reducing relational engagement, effectively becoming the de facto care provider themselves. Therefore attention needs to be paid to caseload capacity and performance monitoring. Caseloads of up to 80 high-intensity service users (Department of Health, 2005b) are likely to be unrealistic, if case managers are to maintain the links that are necessary to communicate

effectively, share information, develop common understandings, establish case reviews and provide the monitoring and back up that is necessary to ensure that plans are followed through (Sargent *et al.* 2008). Gittel (2009) recommends that caseload capacity is assessed in conjunction with cross-functional systems for measuring task integration and operational effectiveness. The advantage of this type of approach is that it factors in the resources that need to be invested in order to improve efficiency and effectiveness of care co-ordination.

This formative evaluation has considerable limitations. It is essentially a single case study and the findings may not be generalisable to other services. The quality of the ACIC survey was impaired by the relatively low response rate and the incompleteness of the dataset. The interview data were also abstracted from long conversations with front line practitioners and focussed on issues that were pertinent to them, rather than broader structural influences that may have exerted an equally influential, though less obvious, effect upon the co-ordination of care. It may have understated the role that institutional rules, regulations, governance structures, information technology and occupational hierarchies play in controlling access to resources and facilitating communication across networks. The analysis highlighted some of the relational mechanisms and organisational influences that played a role in shaping the patterns of care co-ordination. However, more work needs to be carried out in order to explore the subtle and variegated networks of practice that may influence patterns of care co-ordination.

Nevertheless, given that case management for high-intensity service users has received comparatively little attention, it is hoped that this evaluation may stimulate further professional interest and research. Developing task focussed relational workspaces that are systemically managed and adequately resourced, may help to enhance the quality of care co-ordination, particularly in dispersed networks. The hypothetical links that need to be established to move towards this relational form of care co-ordination are summarised in the context, mechanism, outcome configuration CMO – 3 (see Figure 6 below).

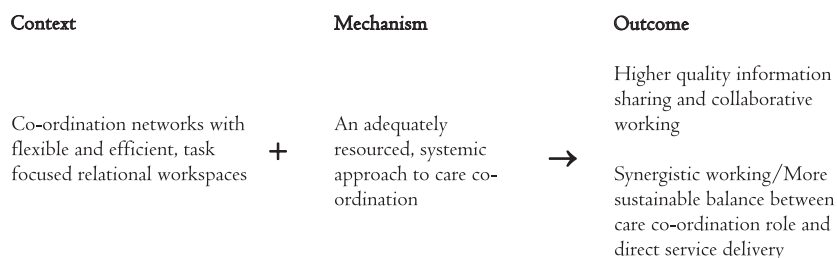


Figure 6 CMO – 3: hypothetical context, mechanism, outcome configuration.

Concluding remarks

Edmondson *et al.* (2001) have suggested that 'when a new technology disrupts existing work routines, the adopting organization must go through a learning process, making cognitive, interpersonal, and organisational adjustments that allow new routines to become ongoing practice (p. 686)'. Case management for high-intensity service users is a relatively new technology. It is hoped that the issues that have been highlighted may help to inform the development of other services that are wrestling with similar challenges. A key issue that may need to be addressed is the need to develop flexible and efficient, task focussed relational workspaces, which support care co-ordination by promoting strong working relationships between co-workers with different functional roles.

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